# Performance Evaluation of FlashAttention-2 for Inference on CPU

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## Abstract

FlashAttention has revolutionized transformer training and inference on GPU by minimizing memory I/O, yet its behavior on CPU remains underexplored. This work provides a performance evaluation of the FlashAttention-2 (FA-2) forward kernel for inference on a single CPU. We benchmark several implementations of the forward kernel against standard attention baselines and, for masked attention, compare three threading strategies for parallelizing the kernel. We also profile the runtime breakdown of the kernel's algorithmic steps. With appropriate threading and block sizes, FA-2 outperforms standard attention in every tested configuration, yielding up to 7.9× speedup for self-attention and 12.8× for masked attention. We also find three intriguing results: larger block sizes can degrade performance despite lower I/O complexity, the optimal block size is typically small-to-moderate, and online softmax dominates runtime despite its lower complexity than matrix multiplications. Our analysis serves as a preliminary exploration of efficient attention computations in resource-constrained settings, including CPU-only inference, and identifies online softmax as a primary optimization target.

## 1. Introduction

The attention mechanism is the core of transformer models (Vaswani et al., 2017), driving their success across various tasks. However, standard attention incurs quadratic computational and I/O complexity with respect to sequence length. As the sequence length grows, the intermediate results of attention often exceed the capacity of fast on-chip memory (SRAM or cache), and the resulting traffic to off-chip memory becomes a major performance bottleneck.

To address this bottleneck, FlashAttention-2 (FA-2) (Dao, 2024) was introduced as a hardware-aware, exact attention algorithm. Through tiling and kernel fusion, it computes attention in a single pass without materializing the large intermediate matrices. While FA-2 has been extensively optimized and adopted for GPU, its performance on general-purpose CPU remains largely unexplored. This gap is significant because CPU inference remains critical for resource-constrained environments (Jiang et al., 2023), such as edge computing, where high-end GPUs are often unavailable.

In this work, we investigate the efficacy of the FA-2 forward kernel for inference on a single CPU. To our knowledge, this is the first systematic empirical study of FA-2 in this setting. We benchmark single- and multi-threaded implementations of the forward kernel against standard attention baselines across a range of sequence lengths, head dimensions, and block sizes. For masked attention, we further evaluate three threading strategies—static, centralized dynamic, and work-stealing—for parallelizing the kernel. We also profile the runtime distribution of the kernel's algorithmic steps.

Our results show that, with suitable threading and block sizes, FA-2 outperforms standard attention across all tested configurations, reaching up to 7.9× speedup for self-attention and 12.8× for masked attention. We further find that the optimal block size is typically small-to-moderate, contradicting the I/O-complexity prediction, and that online softmax accounts for a substantial share of runtime despite its lower theoretical complexity than the matrix multiplications, which is a concrete optimization target. Our code is available for reproducibility.[1]

## 2. Background

### 2.1. Efficient Attention Computations

To compute attention efficiently on long contexts, numerous methods have been proposed to approximate attention (Beltagy et al., 2020; Chen et al., 2021; Choromanski et al., 2021; Katharopoulos et al., 2020; Kitaev et al., 2020; Roy et al., 2021; Zaheer et al., 2020). These encompass sparse attention patterns (Beltagy et al., 2020; Kitaev et al., 2020; Roy et al., 2021; Zaheer et al., 2020), low-rank approximations (Choromanski et al., 2021; Katharopoulos et al., 2020), and hybrid approaches (Chen et al., 2021). On CPU

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*Independent Project.*

[1]https://github.com/Yu-TingLee/flashattn2-cpu

specifically, IceFormer (Mao et al., 2024) and NoMAD-attention (Zhang et al., 2024) accelerate attention inference via approximate methods, using $k$-nearest-neighbor search and SIMD in-register lookups, respectively. However, these approximation methods typically trade model accuracy for theoretical efficiency and may fail to achieve true wall-clock speedups against standard attention.

In contrast, exact attention methods compute the mathematically equivalent attention matrix while optimizing data movement. Specifically, FlashAttention (Dao et al., 2022) and FlashAttention-2 (Dao, 2024) demonstrated that attention is bottlenecked by I/O cost, and attention computations can be substantially faster on GPU by minimizing memory accesses. Crucially, whether these hardware-aware advantages translate to CPU remains an important open question.

### 2.2. Hardware Characteristics: CPU vs. GPU

#### 2.2.1. Performance Characteristics

GPU consists of compute elements (e.g., CUDA cores) and a memory hierarchy. The memory hierarchy comprises high bandwidth memory (HBM) and on-chip SRAM. For example, the A100 GPU has 40-80 GB of HBM with bandwidth 1.5-2.0 TB/s and 192 KB of on-chip SRAM per streaming multiprocessor (108 total) with bandwidth estimated around 19 TB/s (Dao et al., 2022). CPU relies on a multi-level cache hierarchy (L1, L2, L3) backed by off-chip DRAM, where caches are implicitly managed by hardware. As a reference, the i9-11900K used in our experiments has 512 KiB of L2 cache per core and 16 MiB of shared L3 cache, with a peak DRAM bandwidth of approximately 38.4 GB/s.

#### 2.2.2. Execution Model

GPU achieves massive parallelism via the SIMT execution model, where threads execute in groups of 32 (warps) across streaming multiprocessors. Modern GPU also features specialized matrix multiply units (e.g., Tensor Cores) that can deliver over an order of magnitude higher throughput than general-purpose floating-point units. For instance, the A100 achieves 312 TFLOPs/s for FP16/BF16 matmul versus 19.5 TFLOPs/s for non-matmul FP32. In contrast, CPU relies on a smaller number of cores optimized for complex instruction streams, with parallelism expressed through both SIMD vector units and thread-level concurrency.

## 3. A Comparison between Standard Attention and the FA-2 Forward Kernel

Let $W_Q \in \mathbb{R}^{d\times d}, W_K \in \mathbb{R}^{d\times d}, W_V \in \mathbb{R}^{d\times d}$ be the trainable weights and $\widetilde{Z} \in \mathbb{R}^{T\times d}$ be the input matrix. For a transformer model, $T$ specifies the input sequence length and $d$ is the head dimension. The query, key, and value

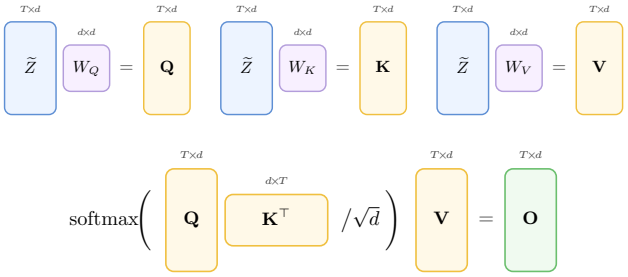



*Figure 1.* **Computation of standard attention.** Input matrix $\widetilde{Z}$ is projected by trainable weights $W_Q$, $W_K$, and $W_V$ to obtain the query matrix $\mathbf{Q}$, key matrix $\mathbf{K}$, and value matrix $\mathbf{V}$. Output matrix $\mathbf{O}$ is obtained by applying row-wise softmax to the scaled product $\mathbf{Q}\mathbf{K}^\top/\sqrt{d}$, then multiplying by $\mathbf{V}$.

matrices are defined to be $\mathbf{Q} := \widetilde{Z}W_Q \in \mathbb{R}^{T\times d}, \mathbf{K} := \widetilde{Z}W_K \in \mathbb{R}^{T\times d}, \mathbf{V} := \widetilde{Z}W_V \in \mathbb{R}^{T\times d}$. The computation of attention can be summarized as

$$\mathbf{O} := \mathrm{softmax}(\mathbf{Q}\mathbf{K}^\top/\sqrt{d})\mathbf{V} \in \mathbb{R}^{T\times d} \tag{1}$$

where $\mathbf{O}$ is the output matrix and the softmax is applied row-wise. See Figure. 1 for a visualization of standard attention.

It is well-known that the computational complexity of standard attention is $\mathcal{O}(T^2d)$, while the I/O complexity (memory accesses) is $\mathcal{O}(T^2)$, when the intermediate $T \times T$ matrices cannot fit in on-chip memory. Therefore, standard attention is less efficient for long-sequence workloads. For a detailed derivation, please see Sec. A. It is thus crucial to design I/O-efficient attention algorithms that reduce memory accesses while preserving computational equivalence to standard attention. Specifically, it is desirable that the high speed of on-chip SRAM can be fully utilized.

A natural approach is to partition $\mathbf{Q}$, $\mathbf{K}$ and $\mathbf{V}$ into row blocks $\mathbf{Q}_{I_1}, \ldots, \mathbf{Q}_{I_{T_r}}$, $\mathbf{K}_{J_1}, \ldots, \mathbf{K}_{J_{T_c}}$, and $\mathbf{V}_{J_1}, \ldots, \mathbf{V}_{J_{T_c}}$ that fit entirely in on-chip memory and to compute the corresponding output row blocks $\mathbf{O}_{I_1}, \ldots, \mathbf{O}_{I_{T_r}}$. This technique is commonly referred to as *tiling* in literature. Tiling alone, however, is not enough.

For example, a safe softmax naively requires three passes over the input: one to compute the row-wise maximum, one to compute the sum of exponentials, and one to compute the normalized output, and these redundant memory accesses degrade performance for long sequences. Moreover, the non-linearity of softmax prevents us from naively summing partial results across column blocks. We need to rescale existing partial output to account for the new normalization factor before adding the contribution from the current blocks of $\mathbf{K}^\top$ and $\mathbf{V}$. Resolving these issues requires either two passes over the data or, as in the FA-2 forward kernel, an *online update* that maintains running statistics and rescales partial outputs as new row-blocks of $\mathbf{K}_{J_j}^\top$ and

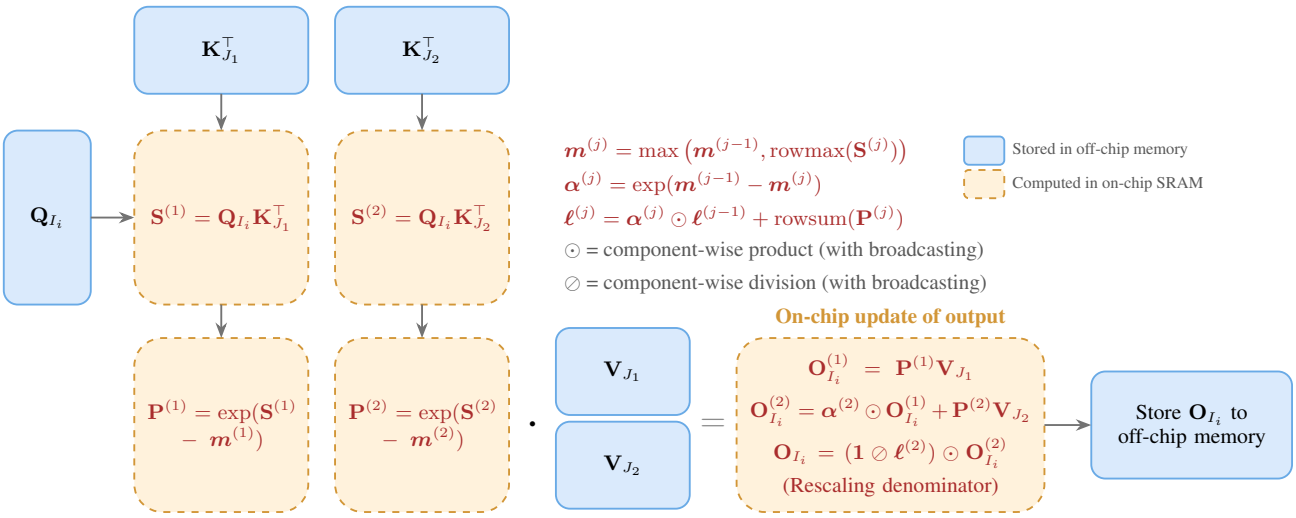



*Figure 2.* **Computation of the FA-2 forward kernel (Algorithm 1).** We assume only $T_c = 2$ column blocks. For a fixed row block $I_i$, the key $\mathbf{K}_{J_j}$ and value $\mathbf{V}_{J_j}$ blocks are loaded sequentially into on-chip SRAM, where the intermediate matrices $\mathbf{S}^{(j)}$, $\mathbf{P}^{(j)}$, and the running statistics $\boldsymbol{m}^{(j)}$, $\boldsymbol{\ell}^{(j)}$ are computed and used to update $\mathbf{O}_{I_i}$ in place. After all column blocks are processed, the output is rescaled by $1/\boldsymbol{\ell}$ and stored to off-chip memory. This avoids materializing the $T \times T$ intermediate matrices $\mathbf{S}$ and $\mathbf{P}$ in off-chip memory.

$\mathbf{V}_{J_j}$ are processed (Sec. B).

The resulting FA-2 forward kernel is shown in Algorithm 1 with a visual illustration in Figure. 2. We can show that Algorithm 1 achieves the same computational complexity of $\mathcal{O}(T^2d)$ while reducing I/O complexity to $\mathcal{O}(\frac{T^2d^2}{M})$, where $M$ is the on-chip memory budget (Sec. C). For typical values of $T$ and $d$, this represents a significant reduction in I/O complexity compared to standard attention. A line-by-line complexity analysis (Sec. C.3) further shows that Line 7 (Load $\mathbf{K}, \mathbf{V}$) dominates the memory accesses with $\mathcal{O}(\frac{T^2d}{|I|})$ cost, while Line 8 (Compute $\mathbf{S}$) and Line 11 (Compute $\mathbf{O}$ in inner loop) dominate the computation complexity, both by the scale of $T^2d$. This analysis also serves as the reference for our profiling experiments in Sec. 5.2.

**Algorithm 1** FA-2 Forward Kernel

1: Divide $\mathbf{Q}$ into row-block matrices $\mathbf{Q}_{I_1}, \dots \mathbf{Q}_{I_{T_r}}$ of size $|I| \times d$ and divide $\mathbf{K}, \mathbf{V}$ into $\mathbf{K}_{J_1}, \dots, \mathbf{K}_{J_{T_c}}$ and $\mathbf{V}_{J_1}, \dots, \mathbf{V}_{J_{T_c}}$ of size $|J| \times d$ each.
2: Divide the output $\mathbf{O}$ into row-block matrices $\mathbf{O}_{I_1}, \dots \mathbf{O}_{I_{T_r}}$ of size $|I| \times d$ each.
3: **for** $i = 1$ to $T_r$ **do**
4: Load $\mathbf{Q}_{I_i}$ to on-chip SRAM.
5: Initialize $\mathbf{O}_{I_i} \leftarrow \mathbf{0}_{|I_i| \times d}$, $\boldsymbol{\ell} \leftarrow \mathbf{0}_{|I_i|}$ and $\boldsymbol{m} \leftarrow (-\infty)_{|I_i|}$ on chip.
6: **for** $j = 1$ to $T_c$ **do**
7: Load $\mathbf{K}_{J_j}$ and $\mathbf{V}_{J_j}$ to on-chip SRAM.
8: Compute $\mathbf{S} \leftarrow \mathbf{Q}_{I_i}\mathbf{K}_{J_j}^\top$ on chip.
9: # $\oslash$ = component-wise division; $\odot$ = component-wise product
10: Compute $\boldsymbol{m}_{\text{new}} \leftarrow \max(\boldsymbol{m}, \text{rowmax}(\mathbf{S}))$, $\mathbf{P} \leftarrow \exp(\mathbf{S} - \boldsymbol{m}_{\text{new}})$, $\boldsymbol{\ell}_{\text{new}} \leftarrow \boldsymbol{\alpha} \odot \boldsymbol{\ell} + \text{rowsum}(\mathbf{P})$ on chip, where $\boldsymbol{\alpha} = \exp(\boldsymbol{m} - \boldsymbol{m}_{\text{new}})$.
11: Compute $\mathbf{O}_{I_i} \leftarrow \boldsymbol{\alpha} \odot \mathbf{O}_{I_i} + \mathbf{P}\mathbf{V}_{J_j}$ on chip.
12: Compute $\boldsymbol{m} \leftarrow \boldsymbol{m}_{\text{new}}, \boldsymbol{\ell} \leftarrow \boldsymbol{\ell}_{\text{new}}$ on chip.
13: **end for**
14: Compute $\mathbf{O}_{I_i} \leftarrow (\mathbf{1} \oslash \boldsymbol{\ell}) \odot \mathbf{O}_{I_i}$ on chip.
15: Store $\mathbf{O}_{I_i}$ to off-chip memory.
16: **end for**

## 4. Experimental Settings

*Table 1.* **CPU specifications.**

| CPU Info | |
|---|---|
| **Model Name** | 11th Gen Intel(R) Core(TM) i9-11900K @ 3.50GHz |
| **Cores / Threads** | 8 Physical / 16 Logical |
| **L2 Cache Size** | 512 KiB per core |
| **L3 Cache Size** | 16 MiB (Shared) |

We evaluate the forward kernel in isolation using 50 reference test sets of synthetic inputs. The reference test sets contain inputs $\mathbf{Q}, \mathbf{K}, \mathbf{V}$ and ground-truth outputs $\mathbf{O}$ used to verify the correctness of all implementations. As end-to-end performance depends on broader system factors like KV cache management and memory allocation overhead, we leave further exploration to future work.

All implementations are written in C++ using Eigen3, which

is a high-performance library for linear algebra, and are compiled with `-O3`, `-march=native`, and `-ffast-math`. CPU specifications are summarized in Table 1.

For self-attention (even workload), we evaluate four variants: single-threaded FA-2 (ST), multi-threaded FA-2 (MT, 8 threads), multi-threaded FA-2 with simultaneous multithreading (MT, 16 threads), and a standard attention baseline. For masked attention (uneven workload), we further compare three scheduling strategies for the multi-threaded variants: (1) static scheduling, (2) centralized dynamic scheduling, and (3) work-stealing scheduling. Alongside the single-threaded masked FA-2 (ST) and the masked attention baseline, we have a total of 8 variants for masked attention.

We measure the runtime of each implementation across sequence lengths $T \in \{256, 512, 1024, 2048, 4096, 8192\}$ and head dimensions $d \in \{64, 128\}$. We manually set the cache budget $M$ to simulate different on-chip SRAM sizes, sweeping values from 32 KiB to 1 MiB. Note that $M$ controls the tiled block size. Specifically, with $M$ in bytes, we compute the row block size $|I|$ and column block size $|J|$ as: $|I| = \lfloor M/(16d) \rfloor, |J| = \min(|I|, d)$. See table 2 for the number of row blocks of representative configurations. We additionally profile the single-threaded implementation on $T \in \{512, 2048, 8192\}$, $d \in \{64, 128\}$, and $M \in \{16\,\text{KiB}, 256\,\text{KiB}, 1\,\text{MiB}\}$ to break down the runtime of specific lines in Algorithm 1. All results are aggregated over the 50 reference test sets.

*Table 2.* **Representative number of row blocks ($T_r$).**

(a) Head dimension $d = 64$

| $M$ (KiB) | $T = 256$ | $T = 512$ | $T = 1024$ | $T = 2048$ | $T = 4096$ | $T = 8192$ |
|---|---|---|---|---|---|---|
| 32 | 8 | 16 | 32 | 64 | 128 | 256 |
| 256 | 1 | 2 | 4 | 8 | 16 | 32 |
| 1024 | 1 | 1 | 1 | 2 | 4 | 8 |

(b) Head dimension $d = 128$

| $M$(KiB) | $T = 256$ | $T = 512$ | $T = 1024$ | $T = 2048$ | $T = 4096$ | $T = 8192$ |
|---|---|---|---|---|---|---|
| 32 | 16 | 32 | 64 | 128 | 256 | 512 |
| 256 | 2 | 4 | 8 | 16 | 32 | 64 |
| 1024 | 1 | 1 | 2 | 4 | 8 | 16 |

## 5. Experimental Results

In this section, we present our experimental results on runtime performance and the profiling analysis.

### 5.1. Runtime of Attention across Sequence Lengths $T$, Head Dimensions $d$, and Cache Budgets $M$

Figure 3 presents the relative speedup compared to standard attention across different sequence lengths $T$, head dimensions $d$, and cache budgets $M$. Our results reveal several performance trends and, notably, contradict the I/O-complexity $\mathcal{O}(\frac{T^2 d^2}{M})$ that larger $M$ should reduce runtime.

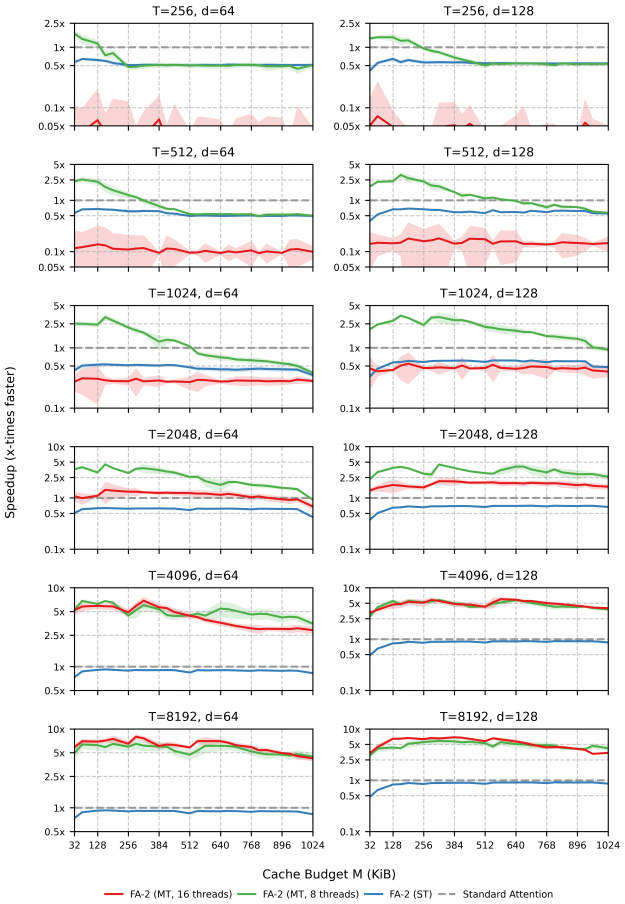



*Figure 3.* **Speedup of the FA-2 forward kernel over standard attention.** Curves represent mean speedup and shaded area represents standard deviation. With appropriate block sizes, FA-2 (MT, 8 threads) is the optimal choice across all $T \leq 2048$ and $d$, while FA-2 (MT, 16 threads) is the best for $T > 2048$ and can reach up to 7.9× speedup when $T = 8192$, $d = 64$, and $M = 288$ KiB.

For short sequences ($T \leq 512$), FA-2 (MT, 8 threads) is the fastest variant only at small block sizes, where it can reach about 2× speedup. As $M$ grows, its speedup decays and eventually drops to FA-2 (ST). FA-2 (MT, 16 threads) performs the worst throughout this regime. This can be explained by the fact that, at small $T$, increasing $M$ rapidly collapses the number of rows toward 1 (Table 2), leaving too few row blocks for parallelization. For medium sequences ($512 < T \leq 2048$), FA-2 (MT, 8 threads) typically outperforms all other implementations, regardless of the block size. For long sequences ($T > 2048$), speedup becomes much less sensitive to block sizes, and the 16-thread variant catches up to and slightly exceeds the 8-thread variant, peaking at 7.9× when $T = 8192$, $d = 64$, and $M = 288$ KiB.

We also note that single-threaded FA-2 (ST) falls behind

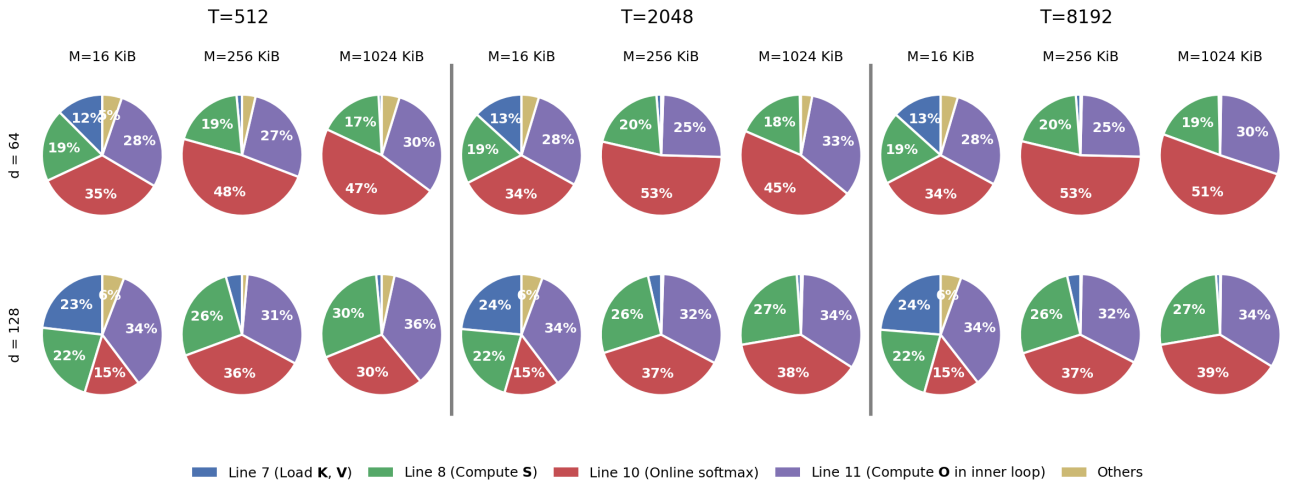



*Figure 4.* **Per-step runtime profiling of the single-threaded FA-2 forward kernel.** Columns group $M$ within each $T$; rows correspond to $d$. We note that Line 10 (Online softmax) consistently consumes the largest single share of runtime despite its lower complexity than the matmuls in Lines 8 and 11.

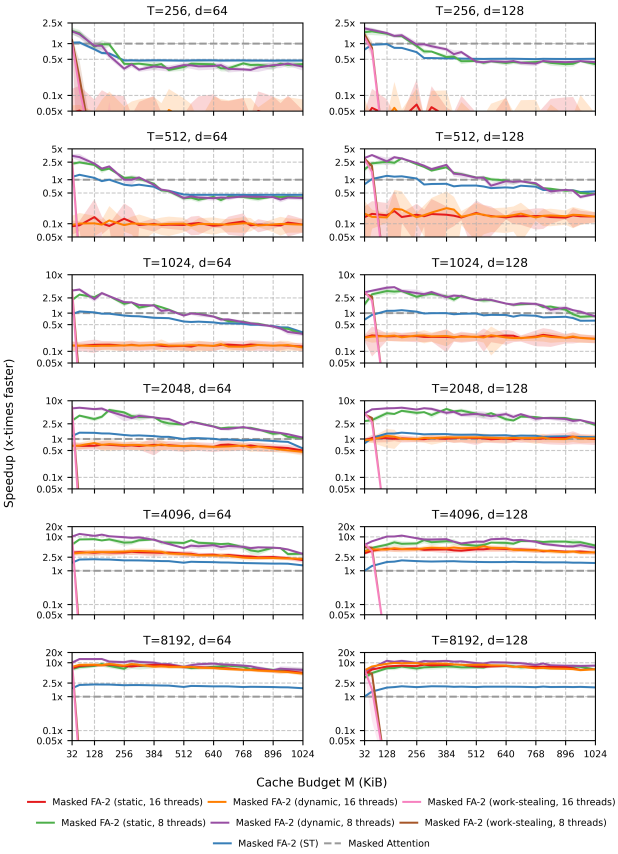
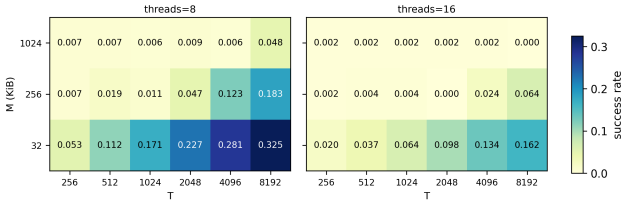
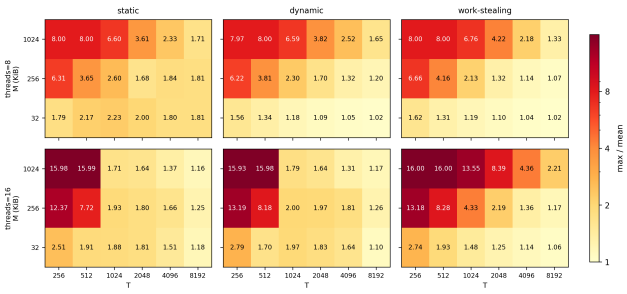



*Figure 5.* **Speedup of masked FA-2 forward kernel over masked attention.** Dynamic scheduling is almost always the best or tied-for-best choice, peaking at 12.8× when $T = 8192$, $d = 64$, and $M = 160$ KiB. Work-stealing collapses sharply with increasing $M$ across all tested configurations.




*(a)* Steal success rate for work-stealing.




*(b)* Load imbalance across static, dynamic, and work-stealing strategies.

*Figure 6.* **Performance metrics of masked attention.** (a) Steal success rate of work-stealing. (b) Load imbalance (max / mean thread runtime) for static, dynamic, and work-stealing. Work-stealing's load imbalance is comparable to dynamic's, yet its runtime is far worse (Figure 5). The steal success rate also collapses with $M$, indicating that the cost of failed steal attempts may drive the performance degradation.

standard attention across all configurations. The algorithmic I/O savings of the forward kernel alone are insufficient to outperform a well-optimized standard attention on CPU; the gains only come from coupling FA-2 with multithreading at appropriate block sizes.

Finally, increasing the head dimension $d$ is typically beneficial for the FA-2 implementations, as larger $d$ increases the per-block arithmetic intensity, which favors FA-2's tiled execution.

### 5.2. Runtime Distribution of Algorithmic Steps

Figure 4 presents the runtime breakdown of the major algorithmic steps. We observe several distinct behaviors dependent on the block size $M$.

At the smallest block size ($M = 16$ KiB), we observe a noticeable share of Others (yellow). This is expected as small tile sizes increase the number of loop iterations and thus the loop overhead becomes more significant. This share shrinks substantially at $M = 256$ and $M = 1024$ KiB. Although Line 7 (Load $\mathbf{K}, \mathbf{V}$) dominates the theoretical I/O (Sec. C.3), its runtime share remains small and becomes negligible when $M = 256$ and $M = 1024$ KiB.

More intriguingly, Line 10 (Online softmax) consistently consumes the largest single share of runtime across nearly every configuration, despite having lower complexity than the matmuls in Lines 8 and 11. This bottleneck is more pronounced when $d = 64$, which is consistent with the line-by-line analysis in Sec. C.3: the matmul costs in Lines 8 and 11 scale with $d$ while Line 10's cost does not, so its relative share grows as $d$ shrinks. Further investigation of this bottleneck is left to future work.

### 5.3. Runtime of Masked Attention across Sequence Lengths $T$, Head Dimensions $d$, and Cache Budgets $M$

Figure 5 presents the speedup of masked FA-2 over the masked attention baseline, and Figure 6 presents the steal success rate of work-stealing and the load imbalance (max / mean thread runtime) for all three strategies.

For masked attention, some trends match those of self-attention: (1) for short to medium sequences ($T \leq 1024$), the best variant, Masked FA-2 (dynamic, 8 threads), wins only at small $M$ (around 2-3×); (2) for $T > 1024$, speedup becomes less sensitive to block size and stays high across various $M$, peaking up to 12.8× by Masked FA-2 (dynamic, 8 threads) when $T = 8192$, $d = 64$, and $M = 160$ KiB;

Among the three strategies, dynamic scheduling is the most robust choice, and is almost always the best or tied-for-best choice. Static tracks dynamic closely but only slightly falls behind at small block size. Work-stealing, in contrast, degrades sharply with $M$. It is competitive only at small $M$: both the 16-thread and 8-thread variants collapse below 0.05× for nearly all $M > 64$ KiB. We therefore recommend dynamic scheduling as the default.

In particular, the collapse of work-stealing is not explained by load imbalance. From Figure 6, we observe that: (1) load imbalance decreases in $T$ and increases in $M$ for all strategies, with near-perfect balance at $T = 8192$, $M = 32$ KiB; (2) work-stealing's load imbalance is comparable to dynamic's in most cells, yet its runtime is far worse; (3) the steal success rate peaks at 0.325 and falls below 0.01 as $M$ increases, meaning that the vast majority of steal attempts are wasted. Together, these indicate that work-stealing pays a cost that does not buy useful work, but the precise cause of this failure is left to future work.

Overall, with appropriate threading and block sizes, the FA-2 forward kernel substantially accelerates both self- and masked attention on CPU. The empirical bottleneck of online softmax also points to a concrete optimization target.

## 6. Discussion

Our study has several limitations. First, our implementations prioritize algorithmic transparency over maximal hardware utilization. Without the hand-tuned SIMD vectorization and memory alignment, our results establish baseline algorithmic behavior rather than an upper bound for CPU inference. Second, we only test a single CPU architecture. Therefore, the phenomena observed in our experiments may not generalize to different CPU architectures (e.g., ARM). Finally, our evaluation utilized synthetic data, and it will be an interesting and important direction to further investigate the performance on real edge devices. Despite these limitations, this study serves as a critical preliminary exploration of the algorithm's behavior on CPU.

## 7. Conclusion

In this work, we evaluated the inference performance of the FA-2 forward kernel on CPU and benchmarked single- and multi-threaded implementations against standard attention baselines. Our findings indicate that parallelized FA-2 forward kernel yields substantial performance gains, with up to 7.9× speedup and 12.8× speedup for masked attention, compared to standard attention. We also note that larger block sizes do not always reduce runtime despite lower I/O complexity. Empirical profiling reveals that the online softmax calculation can be a runtime bottleneck, despite having a smaller order of computational complexity compared to matrix multiplications. Our observations underscore the importance of further investigation of efficient attention algorithms on CPU.

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# A. I/O and Computational Complexity of Standard Attention

For simplicity, we assume a simplified two-level memory hierarchy only consisting of on-chip SRAM and off-chip memory. In general, the input sequence length is much larger than the head dimension, i.e., $T \gg d$, and we may assume the intermediate $T \times T$ matrices $\mathbf{QK}^\top$ and $\mathrm{softmax}(\mathbf{QK}^\top/\sqrt{d})$ cannot be stored in on-chip SRAM.

We analyze the I/O and computational complexity of standard attention.

## A.1. I/O Complexity

We consider the computations of $\mathbf{QK}^\top \in \mathbb{R}^{T\times T}$, $\mathrm{softmax}(\mathbf{QK}^\top/\sqrt{d}) \in \mathbb{R}^{T\times T}$, and $\mathrm{softmax}(\mathbf{QK}^\top/\sqrt{d})\mathbf{V} \in \mathbb{R}^{T\times d}$ and the corresponding major memory accesses. To compute attention, we must sequentially:

- Write $T^2$ elements to off-chip memory to store $\mathbf{QK}^\top$.
- Read $T^2$ elements by loading $\mathbf{QK}^\top$ from off-chip memory to calculate $\mathrm{softmax}(\mathbf{QK}^\top/\sqrt{d})$, and write $T^2$ elements back to off-chip memory.
- Read $T^2$ elements by loading $\mathrm{softmax}(\mathbf{QK}^\top/\sqrt{d})$ from off-chip memory to calculate $\mathrm{softmax}(\mathbf{QK}^\top/\sqrt{d})\mathbf{V}$.

We conclude that standard attention induces $4T^2 = \mathcal{O}(T^2)$ memory accesses.

## A.2. Computational Complexity

On the other hand, the computational complexity for each operation is, respectively,

- $\mathbf{QK}^\top$: $2(TdT) = 2dT^2$,
- $\mathrm{softmax}(\mathbf{QK}^\top/\sqrt{d})$: $4T^2$ (each row takes $4T$ operations for exponentiation, summation, and normalization),
- $\mathrm{softmax}(\mathbf{QK}^\top/\sqrt{d})\mathbf{V}$: $2(TdT) = 2dT^2$.

Summing up, standard attention induces $(4dT^2 + 4T^2) = \mathcal{O}(T^2d)$ operations.

# B. Derivation of the FA-2 Forward Kernel

In this section, we present a derivation of the FA-2 forward kernel (Algorithm 1) and analyze its I/O and computational complexity.

## B.1. Safe Softmax

We first review safe softmax, which is a numerically stable way to compute softmax. Recall that softmax exponentializes each element of a vector $\mathbf{z}$ and normalizes them by the sum of exponentials. This creates numerical instability when some elements of $\mathbf{z}$ are large, leading to overflow during exponentiation. For example, float32 can represent numbers up to approximately $3.4 \times 10^{38}$, which means $\exp(89) \approx 8.6 \times 10^{38}$ already overflows. To mitigate this, safe softmax simply subtracts the maximum element from $\mathbf{z}$ before exponentiation:

$$\frac{\exp(z_i)}{\sum_j \exp(z_j)} = \frac{\exp(z_i - \max_k z_k)}{\sum_j \exp(z_j - \max_k z_k)}. \tag{2}$$

Note that the equality holds trivially as $\exp(z_i - \max_k z_k) = \exp(z_i)/\exp(\max_k z_k)$.

### B.2. Two-Pass Tiled Attention

We now present a derivation of the two-pass tiled attention algorithm, which is the basis for the FA-2 forward kernel. First, we partition $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ into the following row blocks:

$$\mathbf{Q} = \begin{bmatrix} Q_{I_1} \\ Q_{I_2} \\ \vdots \\ Q_{I_{T_r}} \end{bmatrix}, \quad \mathbf{K} = \begin{bmatrix} K_{J_1} \\ K_{J_2} \\ \vdots \\ K_{J_{T_c}} \end{bmatrix}, \quad \mathbf{V} = \begin{bmatrix} V_{J_1} \\ V_{J_2} \\ \vdots \\ V_{J_{T_c}} \end{bmatrix}$$

where $|I_i| = T/T_r =: |I|$ for all $i = 1, 2, \ldots, T_r$ and $|J_j| = T/T_c =: |J|$ for all $j = 1, 2, \ldots, T_c$. For each block $I_i$, we may rewrite attention as

$$\text{softmax}\left(\frac{\mathbf{Q}_{I_i}\mathbf{K}^\top}{\sqrt{d}}\right)\mathbf{V} = \text{softmax}\left(\frac{1}{\sqrt{d}}\left[\mathbf{Q}_{I_i}\mathbf{K}_{J_1}^\top, \ldots, \mathbf{Q}_{I_i}\mathbf{K}_{J_{T_c}}^\top\right]\right)\begin{bmatrix} \mathbf{V}_{J_1} \\ \vdots \\ \mathbf{V}_{J_{T_c}} \end{bmatrix}. \tag{3}$$

Let $\mathbf{z}$ be a row of $\mathbf{Q}_{I_i}\mathbf{K}^\top$. Note that for $j = 1, 2, \ldots, T_c$, the following recurrence holds:

$$m_j = \max(m_{j-1}, \max_{t \in J_j} z_t) \tag{4}$$

$$d_j = d_{j-1} \cdot \exp(m_{j-1} - m_j) + \sum_{t \in J_j} \exp(z_t - m_j) \tag{5}$$

where $m_j = \max_{t \in J_1 \cup \cdots \cup J_j} z_t$ and $d_j = \sum_{t \in J_1 \cup \cdots \cup J_j} \exp(z_t - m_j)$. That is, we can actually compute the maximum and sum of exponentials iteratively over $J_j$. Using the above recurrence and simple matrix algebra, the tiled attention can be computed in two passes (Algorithm 2):

**Algorithm 2** Two-Pass Tiled Attention

Divide $\mathbf{Q}$ into row-block matrices $\mathbf{Q}_{I_1}, \ldots \mathbf{Q}_{I_{T_r}}$ of size $|I| \times d$ and divide $\mathbf{K}$, $\mathbf{V}$ into $\mathbf{K}_{J_1}, \ldots, \mathbf{K}_{J_{T_c}}$ and $\mathbf{V}_{J_1}, \ldots, \mathbf{V}_{J_{T_c}}$ of size $|J| \times d$ each.
Divide the output $\mathbf{O}$ into row-block matrices $\mathbf{O}_{I_1}, \ldots \mathbf{O}_{I_{T_r}}$ of size $|I| \times d$ each.
**for** $i = 1$ to $T_r$ **do**
  Load $\mathbf{Q}_{I_i}$ to on-chip SRAM.
  Initialize $\mathbf{O}_{I_i} \leftarrow \mathbf{0}_{|I_i| \times d}$, $\boldsymbol{\ell} \leftarrow \mathbf{0}_{|I_i|}$ and $\boldsymbol{m} \leftarrow (-\infty)_{|I_i|}$ on chip.
  **for** $j = 1$ to $T_c$ **do**
    Load $\mathbf{K}_{J_j}$ to on-chip SRAM.
    Compute $\mathbf{S}_j \leftarrow \mathbf{Q}_{I_i}\mathbf{K}_{J_j}^\top$ on chip.
    Compute $\boldsymbol{m}_j \leftarrow \max(\boldsymbol{m}_{j-1}, \text{rowmax}(\mathbf{S}_j))$ on chip.
    # $\odot$ = component-wise product
    Compute $\boldsymbol{\ell}_j \leftarrow \exp(\boldsymbol{m}_{j-1} - \boldsymbol{m}_j) \odot \boldsymbol{\ell}_{j-1} + \text{rowsum}(\exp(\mathbf{S}_j - \boldsymbol{m}_j))$ on chip.
    Store $\mathbf{S}_j$ to off-chip memory.
  **end for**
  **for** $j = 1$ to $T_c$ **do**
    Load $\mathbf{S}_j$ and $\mathbf{V}_{J_j}$ to on-chip SRAM.
    # $\oslash$ = component-wise division; $\odot$ = component-wise product
    Compute $\mathbf{P}_j \leftarrow (\mathbf{1} \oslash \boldsymbol{\ell}_{T_c}) \odot \exp(\mathbf{S}_j - \boldsymbol{m}_{T_c})$ on chip.
    Compute $\mathbf{O}_{I_i} \leftarrow \mathbf{O}_{I_i} + \mathbf{P}_j\mathbf{V}_{J_j}$ on chip.
  **end for**
  Store $\mathbf{O}_{I_i}$ to off-chip memory.
**end for**

### B.3. One-Pass Online Attention

While the two-pass approach is an improvement, we can further fuse the passes into a single pass. Let $\mathbf{O}_{I_i}^{(j)} = \sum_{k=1}^{j} \exp(\mathbf{S}_k - \boldsymbol{m}_j)\mathbf{V}_{J_k}$. We observe the following relationship:

$$\mathbf{O}_{I_i}^{(j)} = \left[\sum_{k=1}^{j-1} \exp(\mathbf{S}_k - \boldsymbol{m}_j)\mathbf{V}_{J_k}\right] + \exp(\mathbf{S}_j - \boldsymbol{m}_j)\mathbf{V}_{J_j} \tag{6}$$

$$= \exp(\boldsymbol{m}_{j-1} - \boldsymbol{m}_j) \odot \left[\sum_{k=1}^{j-1} \exp(\mathbf{S}_k - \boldsymbol{m}_{j-1})\mathbf{V}_{J_k}\right] + \exp(\mathbf{S}_j - \boldsymbol{m}_j)\mathbf{V}_{J_j} \tag{7}$$

$$= \exp(\boldsymbol{m}_{j-1} - \boldsymbol{m}_j) \odot \mathbf{O}_{I_i}^{(j-1)} + \exp(\mathbf{S}_j - \boldsymbol{m}_j)\mathbf{V}_{J_j} \tag{8}$$

Now, one sees that

$$\mathbf{O}_{I_i} = (\mathbf{1} \oslash \boldsymbol{\ell}_{T_c}) \odot \mathbf{O}_{I_i}^{(T_c)}. \tag{9}$$

With some algebraic manipulations, we have successfully removed the forward dependency on $\boldsymbol{\ell}_{T_c}$ and $\boldsymbol{m}_{T_c}$. Now, we can fuse the two loops into one and arrive at the FA-2 forward kernel (Algorithm 1).

## C. I/O and Computational Complexity of the FA-2 Forward Kernel

We analyze the I/O and computational complexity of the FA-2 forward kernel.

### C.1. I/O Complexity

For each outer iteration $i$:

- We load $\mathbf{Q}_{I_i}$ and initialize $\mathbf{O}_{I_i}$ once. This costs $\mathcal{O}(|I|d)$.
- We iterate through the inner loop $j = 1, \ldots, T_c$ and load $\mathbf{K}_{J_j}$ and $\mathbf{V}_{J_j}$ for $T_c$ times. This costs $\mathcal{O}((|J| \times T_c)d) = \mathcal{O}(Td)$.
- Finally, we write the result $\mathbf{O}_{I_i}$ back to off-chip memory. This costs $\mathcal{O}(|I|d)$.

Summing over the $T_r$ outer iterations, the number of memory accesses is $\mathcal{O}(T_r \times Td) = \mathcal{O}(\frac{T^2 d}{|I|})$. Choosing $|I| = \Theta(M/d)$, we have total memory accesses $\mathcal{O}(\frac{T^2 d^2}{M})$. For typical values of $T$ and $d$, this is much more I/O efficient.

### C.2. Computational Complexity

For each $(I_i, J_j)$, the matrix products $\mathbf{Q}_{I_i}\mathbf{K}_{J_j}^\top$ and $\mathbf{P}\mathbf{V}_{J_j}$ each costs $\mathcal{O}(|I||J|d)$. Summing over $T_r T_c = \frac{T}{|I|}\frac{T}{|J|}$ blocks, the computational complexity is $\mathcal{O}(T^2 d)$, which is the same as standard attention.

While the above analysis confirms that FA-2 achieves better memory complexity than standard attention, we also provide a finer-grained, line-by-line complexity analysis to accompany our runtime profiling in Sec. 5.2

### C.3. Line-by-Line Complexity Analysis

#### C.3.1. Line-by-Line I/O Complexity

We calculate the exact number of elements transferred between SRAM and off-chip memory.

- **Line 4** (Load $\mathbf{Q}_{I_i}$ to on chip SRAM). Each block $\mathbf{Q}_{I_i}$ contains $|I|d$ elements. So, line 4 yields $Td$ memory access cost.
- **Line 7** (Load $\mathbf{K}_{J_j}$ and $\mathbf{V}_{J_j}$ to on chip SRAM). Each $(i, j)$ iteration reads $2|J|d$ elements. So, line 7 yields $2\frac{T^2 d}{|I|}$ memory access cost.
- **Line 15** (Store $\mathbf{O}_{I_i}$ to off-chip memory). Each block $\mathbf{O}_{I_i}$ contains $|I|d$ elements. So, line 15 yields $Td$ memory access cost.

### C.3.2. Line-by-Line Computational Complexity

Similarly, we break down the exact operation counts for matrix multiplication and online softmax.

- **Line 8** (Compute $\mathbf{S} \leftarrow \mathbf{Q}_{I_i}\mathbf{K}_{J_j}^\top$ on chip). Each $(i, j)$ iteration costs $2|I||J|d$ operations. So, line 8 yields $2T^2d$ total computational cost.
- **Line 10** (Compute $\boldsymbol{m}_{\text{new}}$, $\mathbf{P}$ and $\boldsymbol{\ell}_{\text{new}}$ on chip). Note that $\text{rowmax}(\mathbf{S})$ requires $|I|(|J| - 1)$ operations. Updating $\boldsymbol{m}_{\text{new}}$ requires $|I|$ comparisons. Updating $\mathbf{P}$ requires $2|I||J|$ operations. Updating $\boldsymbol{\alpha} \odot \boldsymbol{\ell}$ requires $3|I|$ operations. Updating $\text{rowsum}(\mathbf{P})$ requires $|I|(|J| - 1)$ operations. Updating $\boldsymbol{\ell}_{\text{new}}$ requires additional $|I|$ additions. Summing up, line 10 requires $4|I||J| + 3|I|$ operations per execution. So, line 10 yields $4T^2 + 3\frac{T^2}{|J|}$ total computational cost.
- **Line 11** (Compute $\mathbf{O}_{I_i} \leftarrow \boldsymbol{\alpha} \odot \mathbf{O}_{I_i} + \mathbf{P}\mathbf{V}_{J_j}$ on chip). Updating $\boldsymbol{\alpha}$ requires $2|I|$ operations. Component-wise product with $\mathbf{O}_{I_i}$ requires $|I|d$ operations; $\mathbf{P}\mathbf{V}_{J_j}$ requires $2|I||J|d$ operations. Updating $\mathbf{O}_{I_i,:}$ requires additional $|I|d$ additions. Summing up, line 11 requires $2|I|(1 + d + |J|d)$ operations per execution. So, line 11 yields $2T^2d + 2T^2\frac{d}{|J|} + 2\frac{T^2}{|J|}$ total computational cost.
- **Line 14** (Compute $\mathbf{O}_{I_i} \leftarrow (\mathbf{1} \oslash \boldsymbol{\ell}) \odot \mathbf{O}_{I_i}$ on chip). Note that $\mathbf{1} \oslash \boldsymbol{\ell}$ requires $|I|$ operations. Component-wise product with $\mathbf{O}_{I_i}$ requires $|I|d$ operations. Summing up, line 14 requires $|I| + |I|d$ operations per execution. So, line 14 yields $Td + T$ total computational cost.